

Week 2 Problem Set

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April 8, 2026

```
#####  
## title: second problem set!  
## author: Jolie Barga  
## purpose: to try out R  
## date: April 8, 2026  
#####
```

On GitHub, create a new **repository** called “soc10problemsets” (or something similar). Then upload the “Week2PSetTemplate.Rmd” file (you can download it here here). Make Tyler ([tylermcdaniel](#)) and Yao ([yao-mxu](#)) “collaborators” in this repository so we can see your work (Settings -> Collaborators -> Add People). Add your name and the date to the top of the file, save your changes, and **commit** them. (note: You can do this online, but I recommend using GitHub Desktop!).

Link to my repository: <https://github.com/joliebarga/soc10problemsets>

2. In the space below, set “eval = TRUE” and try running the first code chunk. Then fix the error. On your local file, **commit** the file to your GitHub.

I fixed the error below.

```
library(dplyr)  
  
## Warning: package 'dplyr' was built under R version 4.5.2  
  
##  
## Attaching package: 'dplyr'  
  
## The following objects are masked from 'package:stats':  
##  
##   filter, lag  
  
## The following objects are masked from 'package:base':  
##  
##   intersect, setdiff, setequal, union  
  
# this is an initial dataframe  
df1 <- data.frame("resource name" = "a",  
                  "custom_ready" = "b")
```

```

# we want to add observations to it (c, and d)
# something is wrong with the following function - can you fix it?
add_line <- function(dataframe){

  new_dataframe <- data.frame("resource.name" = "c",
                              "custom_ready" = "d")

  bind_rows(dataframe, new_dataframe)

}

# this should return a dataframe with all our observations (a, b, c, and d, neatly organized)
add_line(df1)

```

```

##   resource.name custom_ready
## 1             a             b
## 2             c             d

```

3. Add your own data to a collaboration! In the markdown file, add to the small dataframe with some potential data sources for further social science research. Add one or more resources to this list. **Commit** your changes.

```

# write code for question 3 here!
resources_df <- data.frame("resource name" = c("All Our Ideas", "Pew Research Center"),
                           "link" = c("https://allourideas.org/", "https://www.pewresearch.org/datasets/"),
                           "custom_ready" = c("custom made", "previously collected"),
                           "description" = c("All Our Ideas is a wiki survey that asks participants to comp

```

4. Using the `gtrendsR` R package, download data on a term of your choosing. Show a plot of the search term over time, and then plot it over different geographies. Discuss how each of Salganik's 10 characteristics of computational data apply (or don't apply). **Commit** your changes.

My text answers for this question will follow my graphd.

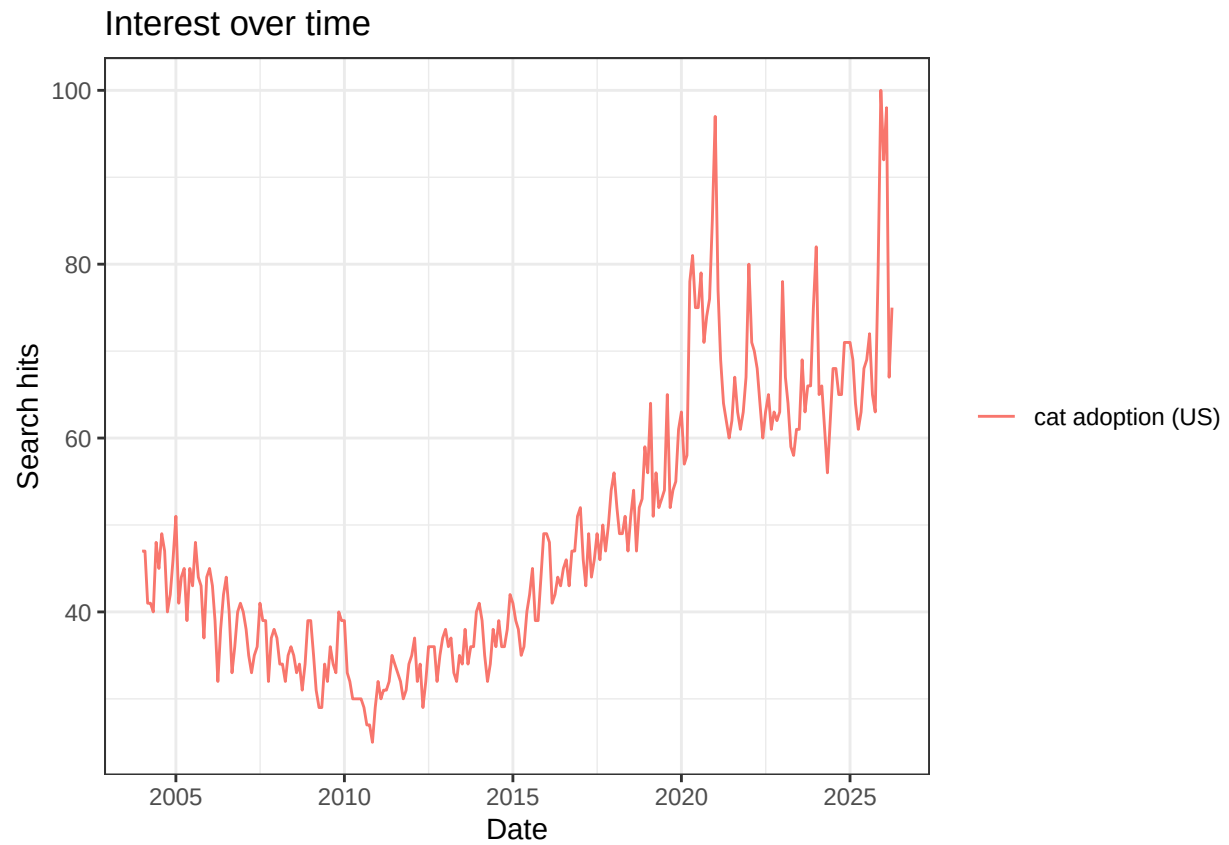
```

library(gtrendsR)
#library(ggplot2)

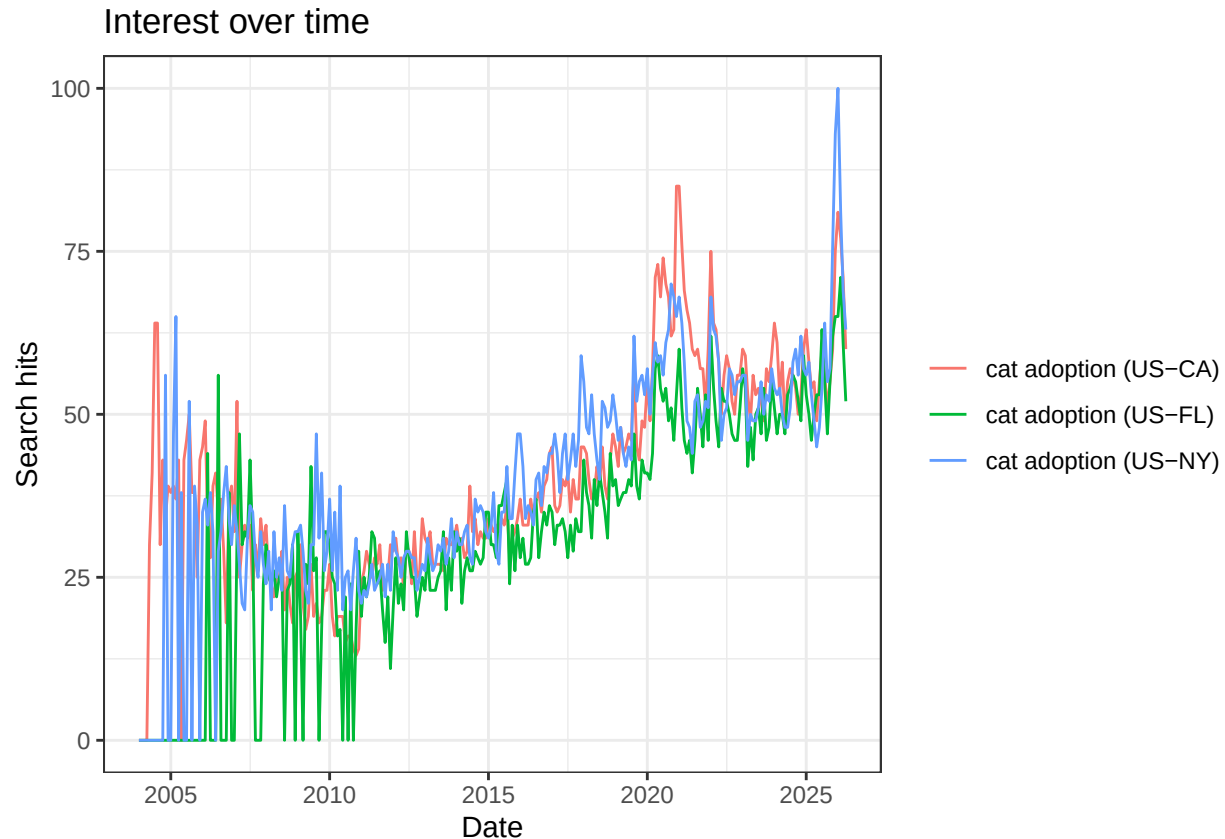
data_social_all <- gtrends(c("cat adoption"),
                           geo = c("US"), time = "all")

plot(data_social_all)

```



```
data_geo <- gtrends(c("cat adoption"),  
                    geo = c("US-CA", "US-NY", "US-FL"), time = "all")  
plot(data_geo)
```



In Bit By Bit, Salganik notes ten characteristics of big data. Here is how each connects (or doesn't connect) to the computational data above on cat adoption:

Big: Yes as google trends is big as it captures data for hundreds of millions of Google users and their search history. **Always-on:** Yes the collection for this data is always on and goes back to collection from 2004 and all google users are included in this data. As shown in the graph there is no time plots in the data so it has been and is continuous. **Non-reactive:** Yes as users aren't aware of their searches being collected so they search 'cat adoption' as they normally do. **Incomplete:** I feel we cant look at this data to understand cat adoptions as a whole because this is simply searching Google for it and this doesn't account for people who go in person to a shelter look and adopt cats, use other applications to search, or who use flyers/other print media to adopt and find adoptable cats. **Inaccessible:** We are given a 0-100 to represent search hits but not an exact count of how many searches as this data is protected so it is to some degree inaccessible to fully understand this search data beyond the 0-100 scale. **Non-representative:** Slightly as using Google search data doesn't represent people without technology, or phones, or dont use Google who adopt and look for adoptable cats or shelters. Older populations are more likely to use newspapers or print adds so this population is likely left out of the demographic. **Drifting:** I do see drifting in the data as their is a large spike in the graph around 2020 during the COVID-19 pandemic therefore representative of a time of surge in pet adoptions as people were solely in their house and did not go to typical in-person work and school schedules. I see a change from pre- to post- Covid where following the pandemic the number of average cat adoption searches remained high showing the data drifted with a real world event. **Algorithmically confounded:** Google does autocomplete searches and give search suggestions so the data may be skewed as people could be pushed towards or away from searching "cat adoption" depending on the algorithm and search index of Google. **Dirty:** Some searches for cat adoption may be from people researching the topic and not to actually adopt cats themselves so the data is dirty because when it comes to representing cat adoptions it could be both people actually interested in cat adoption to adopt cats and for people who just are looking at the topic for academic or other reasons. This contributes to noise in the data and prevents the ability to generalize it and understand the motives behind these searches. **Sensitive:** This is not sensitive because cat adoption is

a standard search and doesn't reveal anything about people from the search.

5. List the three types of collaboration that Salganik names. When can each work well, and what are potential downsides? List one new idea for a collaborative data project in each category. **Commit your changes.**

The three types of collaboration that Salganik names are human computation, open calls, and distributed data collection.

Human computation works well when a task is way too big for a single individual researcher or group but not difficult for volunteers and could even be fun. The example from class was galaxy Zoo where researchers wanted to categorize galaxies but couldn't do it on their own so they used trained volunteers to do this work and were able to produce over 40 million galaxy classifications. It can work well if the task is hard for computers but easier and interesting for people and a downside is that these tasks can only work well if they are simple and repetitive enough for untrained volunteers to be able to complete, so more complex and specialized research where high level training is necessary wouldn't work. A new idea would be if you want to understand public sentiment around the current political climate, you could have volunteers read the public opinion and politics section of the local newspaper to label the emotional tone and feelings and the public to aid sociologists during the current political climate.

Secondly, open call is a type of collaboration where the solution is unknown to researchers but may be known to someone else in the world or by bringing people of specialized knowledge together. The Netflix Prize was a good example of this as they opened it to the public to ask what a good recommendation system was and compared it to their own and if it beat their own they would be able to have good feedback to use a new system. A upside to this is that combining knowledge of people is a great way to work together and collaborate for the greater good, saves time in research, and harnesses people's strengths and talents as well as combines general knowledge for societal advancement. A downside is that there must be a main objective and clear way to evaluate what is brought in and discovered in the open call which can oftentimes be difficult and not apparent in social science research. A new idea would be to ask the public to find the best model and methods for predicting college dropout rates using publicly available education data to find greater correlation between college dropouts and past academic progress.

Finally, the third method of collaboration is distributed data works well when the data researchers need is extremely difficult or impossible to collect on their own, but easy and enjoyable for volunteers to gather as part of their everyday lives. eBird is a good example of this, where birders volunteer information on birds they see in the wild and upload them to a database that helps ornithologists track migration patterns. This is useful when participants can naturally collect data as they go about their routines. A downside is that the data may be inconsistent or unreliable since it is collected by many different people. A new idea could be having volunteers track the number of cats they see on the streets of New York City to estimate the population of stray cats and help researchers better understand patterns in urban animal populations.